

The empirical recurrence for OEIS A224307, recovered from its tail

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Abstract

OEIS A224307 records “Empirical recurrence of order 61” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 280$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 9$, so the recurrence is proved for every $n \geq 70$. Its last coefficient is $c_{61} = 15925248 \neq 0$, so no further tail satisfies anything shorter; any order-61 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A224307 is “Number of $n \times 5$ 0..2 arrays with diagonals and rows unimodal and antidiagonals nondecreasing.”. It has offset 1 and begins

86, 1915, 20214, 164514, 1258372, 9829605, 80083648, 677557164, ...

The entry states:

Empirical recurrence of order 61 (see link above)

As of the “Last modified” line on the live entry (Oct 03 2025 23:21:57, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 280$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 280$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 61: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 9$ is the first whose minimal order is exactly the stated 61.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 560$ exact terms from $n = 9$ on, it returns order 61 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{61} c_i a(n-i),$$

c_1	21	c_{17}	449042010	c_{33}	2144420284826	c_{49}	1321291719072
c_2	−141	c_{18}	−1016438978	c_{34}	1336126755062	c_{50}	67808309792
c_3	225	c_{19}	1583124881	c_{35}	5737546377822	c_{51}	−512574208224
c_4	749	c_{20}	−3025948429	c_{36}	1755325016984	c_{52}	−534137240832
c_5	1466	c_{21}	3907868665	c_{37}	−4587234671076	c_{53}	−272992875264
c_6	−29455	c_{22}	−12792927652	c_{38}	−2472401825642	c_{54}	−133991740416
c_7	70011	c_{23}	−1676823214	c_{39}	−8926411424004	c_{55}	−21331507200
c_8	−12444	c_{24}	−30091474473	c_{40}	−9976114115936	c_{56}	−2906628096
c_9	45390	c_{25}	−4680834856	c_{41}	−4757631819432	c_{57}	3410288640
c_{10}	−1136658	c_{26}	−93247995386	c_{42}	−1582654733788	c_{58}	41140224
c_{11}	3072228	c_{27}	−14823058254	c_{43}	−2835263409452	c_{59}	656916480
c_{12}	−3754535	c_{28}	−354756138691	c_{44}	1998064475416	c_{60}	0
c_{13}	16036928	c_{29}	184796264267	c_{45}	1676569526800	c_{61}	15925248
c_{14}	−61531800	c_{30}	578060428755	c_{46}	2014685415256		
c_{15}	131417385	c_{31}	446206482109	c_{47}	4244926752080		
c_{16}	−161890622	c_{32}	2039947152231	c_{48}	2081184346368		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 70$, and it is the recurrence OEIS A224307 records.*

Proof. That it holds is Lemma ?? applied to the tail from $n = 9$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{61} - c_1 x^{60} - \dots - c_{61}$. Its constant term is $-c_{61} = -15925248$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with

$\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 61 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 61, so $\deg q = 61$, and it is monic. It holds from some index t on. From $\max(t, 9)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 61, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 61, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 15925248.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-61 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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